This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

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P.A. None
Other Cust
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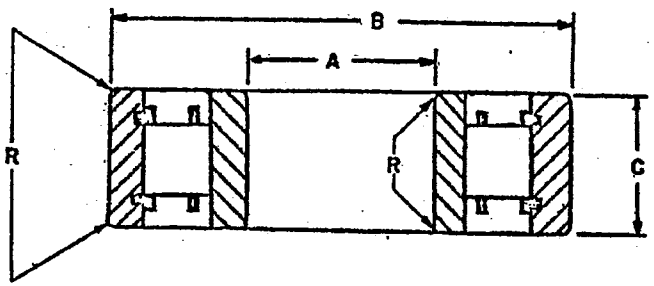
PROCUREMENT SPECIFICATION
Fr-4b-102

SUPERSEDES:

TITLE
BEARINGS, ROLLER, CYLINDRICAL, SINGLE ROW OF ROLLERS, CYLINDRICAL INNER RING, TWO ROLLER RETAINING SNAP RINGS ON OUTER RING, NON-LOCATING.

MILITARY STANDARD
MS 19075

SHEET 1 OF 3



THE CORNER RADIUS OR CHAMFER ON BOTH SIDES OF THE BEARING MUST CLEAR THE SHAFT AND HOUSING FILLET RADIUS LISTED

MS PART NO	FIIN	A INSIDE DIA		a OUTSIDE DIA		C WIDTH		R SHAFT & HOUSING FILLET RADIUS	RECOMMENDED SHOULDER DIAMETER		INTERNAL RADIAL CLEARANCE .0001 INCH		BASIC DYNAMIC LOAD RATING	BASIC STATIC LOAD RATING	RECOMMENDED LIMITING SPEED	EST WEIGHT
									SHAFT	HOUSING						
		MM	INCH	MM	INCH	MM	INCH	MAX	MIN	MAX	MIN	MAX	LBS	LBS	RPM	LBS
MS19075 - 1		17	.6693	40	1.5748	12	.4724	.025	.84	1.44	4	16	2500	1400	7450	.19
MS19075 - 2		17	.6693	47	1.8504	14	.5512	.040	.90	1.67	4	16	2900	1500	8650	.32
MS19075 - 3		20	.7874	52	2.0472	15	.5906	.040	1.02	1.82	4	16	3500	1900	5900	.37
MS19075 - 4		25	.9843	62	2.0472	15	.5906	.040	1.22	1.86	4	18	3000	1600	5500	.35
MS19075 - 5		25	.9843	62	.4409	17	.6693	.040	1.25	2.20	4	18	5100	2900	5100	.64
MS19075 - 6		30	1.1811	62	2.4409	16	.6299	.040	1.42	2.24	4	18	3950	2200	4650	.57
MS19075 - 7		30	1.1811	72	2.8346	19	.7480	.040	1.50	2.55	4	18	6750	3950	4250	.96
MS19075 - 8		35	1.3780	72	2.8346	17	.6693	.040	1.66	2.61	6	20	5900	3450	3950	.75
MS19075 - 9		35	1.3780	80	3.1496	21	.8268	.060	1.73	2.82	6	20	8850	5350	3700	1.29
MS19075 - 10		40	1.5749	80	3.1496	18	.7087	.040	1.87	2.88	5	20	7650	4650	3950	.99
MS19075 - 11		40	1.5748	90	3.5433	23	.9055	.060	1.94	3.19	6	20	10300	6350	3250	1.74
MS19075 - 12		45	1.7717	85	3.3465	19	.7480	.040	2.08	3.06	8	22	8050	5000	3250	1.15
MS19075 - 13		45	1.7717	100	3.9370	25	.9843	.060	2.20	3.54	8	22	14200	9050	2900	2.22
MS19075 - 14		50	1.9685	110	4.3307	27	1.0630	.080	2.42	3.90	8	22	17000	11100	2700	3.04
MS19075 - 15		55	2.1654	100	3.9370	21	.8268	.060	2.52	3.60	8	26	10300	6700	2750	1.71
MS19075 - 16		55	2.1654	120	4.7244	29	1.1417	.080	2.65	4.25	8	26	20700	13600	2400	3.36
MS19075 - 17		60	2.3622	110	4.3307	22	.8661	.060	2.73	3.96	8	26	12600	8450	2500	2.29
MS19075 - 18		60	2.3622	130	5.1181	31	1.2205	.080	2.88	4.61	8	26	23200	15500	2200	4.32
MS19075 - 19		65	2.5591	120	4.7244	23	.9055	.060	3.00	4.32	8	26	14900	10200	2350	2.83
MS19075 - 20		65	2.5591	140	5.5118	33	1.2992	.080	3.11	4.96	8	26	25600	17100	2050	5.99
MS19075 - 21		70	2.7559	125	4.9213	24	.9449	.060	3.19	4.50	10	30	14900	10200	2150	3.02
MS19075 - 22		70	2.7559	150	5.9055	35	1.3780	.080	3.32	5.31	10	30	30100	20800	1900	7.47
MS19075 - 23		75	2.9528	130	5.1181	25	.9843	.060	3.37	4.68	10	30	17000	12100	2350	3.32
MS19075 - 24		75	2.9528	160	6.2992	37	1.4567	.080	3.57	5.67	10	30	33300	25600	1800	9.55
MS19075 - 25		80	3.1496	140	5.5118	26	1.0235	.080	3.60	5.05	10	30	19600	13700	1900	3.32
MS19075 - 26		80	3.1496	170	6.6929	39	1.5354	.080	3.79	6.02	10	30	36500	25600	1650	10.47
MS19075 - 27		85	3.3465	150	5.9055	28	1.1024	.080	3.85	5.40	12	32	22400	15500	1800	5.00
MS19075 - 28		85	3.3465	180	7.0866	41	1.6142	.100	4.04	6.38	12	32	42900	30500	1550	12.26
MS19075 - 29		90	3.5433	160	6.2992	30	1.1811	.080	4.05	5.76	12	32	28000	20200	1650	7.58
MS19075 - 30		90	3.5433	190	7.4803	43	1.6929	.100	4.25	6.73	12	32	46500	33300	1500	14.21
MS19075 - 31		95	3.7402	170	6.6929	32	1.2598	.080	4.29	6.12	12	32	31400	22800	1550	7.16
MS19075 - 32		95	3.7402	200	7.8740	45	1.7717	.100	4.54	7.09	12	32	53300	39100	1400	16.94
MS19075 - 33		100	3.9370	180	7.0866	34	1.2598	.080	4.56	6.48	12	32	34900	25600	1500	8.62

APPROVED 30 APR 1959

REVISED

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MS PART NO	FIIN	A INSIDE DIA MM	A INSIDE DIA INCH	B OUTSIDE DIA MM	B OUTSIDE DIA INCH	C WIDTH MM	C WIDTH INCH	R SHAFT & HOUSING FILLET RADIUS MAX	RECOMMENDED SHOULDER DIAMETER SHAFT MIN	RECOMMENDED SHOULDER DIAMETER HOUSING MAX	INTERNAL RADIAL CLEARANCE .0001 INCH MIN	INTERNAL RADIAL CLEARANCE .0001 INCH MAX	BASIC DYNAMIC LOAD RATING LBS	BASIC STATIC LOAD RATING LBS	RECOMMENDED LIMITING SPEED RPM	EST WEIGHT LBS
MS19075 - 34		100	3.9370	215	8.4646	47	1.8504	.100	4.83	7.62	12	32	61500	45900	1300	19.50
MS19075 - 35		105	4.1339	190	7.4803	36	1.4173	.080	4.77	6.84	14	35	38000	30000	1250	10.32
MS19075 - 36		105	4.1339	225	8.8583	49	1.9291	.100	5.05	7.97	14	35	69700	52400	1250	22.95
MS19075 - 37		110	4.3307	200	7.8740	38	1.4961	.080	5.01	7.20	14	35	45000	34000	1300	11.81
MS19075 - 38		110	4.3307	240	9.4488	50	1.9685	.100	5.37	8.50	14	35	79100	60300	1150	27.38
MS19075 - 39		120	4.7244	215	8.4646	40	1.5748	.080	5.47	7.74	14	35	48500	37500	1280	14.70
MS19075 - 40		120	4.7244	260	10.2362	55	2.1654	.100	5.82	9.21	14	35	93400	71100	9750	36.00
MS19075 - 41		130	5.1181	280	11.0236	58	2.2835	.120	6.32	9.92	16	41	110000	90100	300	43.95
MS19075 - 42		150	5.9055	320	12.5984	65	2.5591	.120	7.11	11.34	16	45	133000	106000	800	60.00

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APPROVED 30 APR 1959 REVISED

P.A. None
Other Cust
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TITLE
BEARINGS, ROLLER, CYLINDRICAL, SINGLE ROW OF ROLLERS, CYLINDRICAL INNER RING, TWO ROLLER RETAINING SNAP RINGS ON OUTER RING, NON-LOCATING.

MILITARY STANDARD
MS 19075

PROCUREMENT SPECIFICATION
FF-B-185

SUPERSEDES

SHEET 2 OF 3

FSC 3110

FORM 672-1

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P.A. None
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PROCUREMENT SPECIFICATION
FF-B-185

SUPERSEDES

TITLE
BEARINGS, ROLLER, CYLINDRICAL, SINGLE ROW OF ROLLERS, CYLINDRICAL INNER RING, TWO ROLLER RETAINING SNAP RINGS ON OUTER RING, NON-LOCATING.

MILITARY STANDARD
MS 19075

SHEET 1 OF 3

572-1

TOLERANCES: STANDARD GRADE (TOLERANCES IN .0001 INCH)

INNER RING				
BORE IN – MM		A		RADIAL RUNOUT
OVER	INCL	MIN	MAX	
0	10	- 3	+0	3
10	18	- 3	+0	4
18	30	- 4	+0	5
30	50	- 5	+0	6
50	80	- 6	+0	8
80	120	- 8	+0	10
120	180	-10	+0	12
180	250	-12	+0	16

OUTER RING				
OD MM		B		RADIAL RUNOUT
OVER	INCL	MAX	MIN	
0	18	+0	- 4	6
18	30	+0	- 4	6
30	50	+0	- 5	8
50	80	+0	- 5	10
80	120	+0	- 6	14
120	150	+0	- 8	16
150	180	+0	-10	18
180	250	+0	-12	20
250	315	+0	-14	24
315	400	+0	-18	28

WIDTH			
NOMINAL BORE MM		C	
OVER	INCL	MAX	MIN
0	180	0	50

THE TOLERANCES FOR WIDTH APPLIES TO INDIVIDUAL RINGS.

MATERIAL: RINGS AND ROLLERS: ALLOY STEEL, FED. STD. 66 STEEL No's., 4320, 4620, 8620 OR E52100. CAGES: CARBON STEEL (ONE PIECE STAMPED).

HARDNESS: RINGS AND ROLLERS: 58 RC. TO 66 RC.

DIMENSIONS: UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.

LUBRICATION: THE IDENTIFYING NUMBER DOES NOT INCLUDE LUBRICATION.

BASIC DYNAMIC LOAD RATING: BASIC DYNAMIC LOAD RATING IS THAT CONSTANT STATIONARY RADIAL LOAD WHICH A GROUP OF APPARENTLY IDENTICAL BEARINGS WITH STATIONARY OUTER RING CAN ENDURE FOR A RATING LIFE OF ONE MILLION REVOLUTIONS OF THE INNER RING.

RATING LIFE (HOURS): RATING LIFE IS THE NUMBER OF HOURS AT SOME CONSTANT SPEED OF THE INNER RING THAT 90 PER CENT OF A GROUP OF APPARENTLY IDENTICAL BEARINGS WILL COMPLETE OR EXCEED BEFORE THE FIRST EVIDENCE OF FATIGUE DEVELOPS.

THE MAGNITUDE OF THE APPROXIMATE RATING LIFE IN HOURS, H_r, IS FOUND FROM THE FORMULA:

$$H = \frac{16667}{R}\left(\frac{C}{P}\right)^{\frac{10}{3}}$$

WHERE C = BASIC DYNAMIC LOAD RATING
P = EQUIVALENT LOAD (COMBINED RADIAL AND SHAFT LOCATION LOAD)
R = REVOLUTIONS PER MINUTE

AVERAGE LIFE IS APPROXIMATELY FIVE TIMES THE RATING LIFE.

BASIC STATIC LOAD RATING: BASIC STATIC LOAD RATING IS THE MAXIMUM STATIC RADIAL LOAD WHICH WILL PRODUCE A PERMANENT DEFORMATION IN THE RACE AND ROLLERS, COMBINED AT ANY ONE CONTACT POINT, OF NOT MORE THAN .0001 OF THE ROLLER DIAMETER.

OPERATING TEMPERATURE: RECOMMENDED OPERATING TEMPERATURE NOT TO EXCEED 250 DEGREES FAHRENHEIT.

LIMITING RPM: LIMITING RPM IS THE RECOMMENDED SAFE DESIGN SPEED FOR NORMAL OPERATING CONDITIONS WITH GREASE LUBRICATION (INNER RING ROTATING).

NOTES:
(A) THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERRED HEREIN.
(B) REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BID.

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